QUANTITATIVE METHODS 1

Worksheet 2 – Measures of central tendency

Exercises

Exercise 1 – Choosing and calculating the appropriate measure of centre

A. Basket values are $18, 20, 20, 22, 25, 95$ euros. Calculate mode, median and arithmetic mean. Which measure best represents a typical customer? Why can the mean still matter?

B. Three grades are 14, 11 and 16, weighted by 30%, 30% and 40%. Calculate the weighted mean.

C. Revenue changes by $+5\%$, -2% and $+12\%$. Calculate the appropriate average annual growth rate.

D. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the appropriate average speed.

Exercise 2 – Grouped distribution

Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5

1. Identify the modal class, taking unequal widths into account.
2. Calculate the median by linear interpolation.
3. Calculate Q_1 and Q_3 by interpolation.
4. Calculate the approximate mean using class midpoints.
5. Find the maximum price such that about 60% of customers can afford at least that price.

Exercise 3 – Which indicator should be used?

For each situation, choose the most appropriate indicator and justify in one sentence: most frequently sold product category; typical salary with a few very highly paid executives; average price when purchased quantities differ; average multi-year growth; average speed over equal distances; centre of a roughly symmetric numerical distribution without extreme values.